E-COMMERCE PLATFORM FOR PERSONALIZED RETAIL EXPERIENCES

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DECLARATION

I declare that this is my own work and this dissertation1 does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

The retail industry is undergoing a fundamental transformation driven by data-intensive personalization strategies. Traditional loyalty programmes rely on static, one-size-fits-all tier structures that fail to capture the nuanced purchasing behaviours of individual customers. This dissertation addresses that gap by designing, implementing, and evaluating a Personalized Retail System that combines unsupervised machine learning, probabilistic modelling, and a novel dynamic loyalty tier framework to deliver context-aware customer experiences.

The proposed system uses the Online Retail II dataset from the UCI Machine Learning Repository as its training corpus. RFM (Recency, Frequency, Monetary) analysis is applied to engineer three core behavioural features. A K-Means clustering model is trained on these features and produces six *parent clusters* that capture broad engagement archetypes: Moderately Engaged Buyers, Inactive or Dormant Buyers, Occasional or Low-Value Buyers, Dormant or Lost Buyers, Low Engagement Buyers, and Recently Engaged High-Value Buyers.

Each parent cluster is further refined by a set of nine static *sub-clusters* drawn from established RFM segmentation literature: Champions, Potential Loyalists, Loyal Customers, Promising, New Customers, Need Attention, About to Sleep, At Risk, and Hibernating. In addition, a probabilistic Naive Bayes model analyses purchase transaction histories to infer per-customer *category preferences*, outputting a preferred category, preference ratio, ranked top categories, and per-category monetary contribution.

The novelty of this work lies in the integration of parent cluster, sub-cluster, and category preference signals into a unified *Behaviour Profile* that drives a four-tier dynamic loyalty programme (Bronze, Silver, Gold, Platinum). Assignment to a loyalty tier is governed by a deterministic scoring function that combines normalised RFM metrics, sub-cluster score, and category preference confidence. Experimental evaluation on 4,338 customers demonstrates that the composite loyalty tier model explains significantly more variance in repeat-purchase revenue than conventional single-variable tier assignment, and that personalised category-aware promotions achieve a simulated redemption lift of 34 % over generic offers.

Keywords: customer segmentation, K-Means clustering, RFM analysis, loyalty tiers, personalized retail, Naive Bayes, behavioral profiling, Online Retail II.

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List of Abbreviations

Abbreviation	Description
API	Application Programming Interface
CRM	Customer Relationship Management
CLV	Customer Lifetime Value
DB	Database
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
EM	Expectation Maximisation
ETL	Extract, Transform, Load
GMM	Gaussian Mixture Model
KPI	Key Performance Indicator
ML	Machine Learning
NB	Naive Bayes
PCA	Principal Component Analysis
RFM	Recency, Frequency, Monetary
ROI	Return on Investment
SLIIT	Sri Lanka Institute of Information Technology
SVM	Support Vector Machine
UCI	University of California, Irvine
UI	User Interface
UX	User Experience
WSS	Within-Cluster Sum of Squares

1. Introduction

1.1 Background

The global retail industry generated revenues exceeding USD 27 trillion in 2023, yet average customer retention rates remain stubbornly below 40 % for most e-commerce operators [1]. In an environment of commoditised products and near-zero switching costs, the ability to understand, predict, and respond to individual customer behaviour has become a primary determinant of competitive advantage.

Customer segmentation—the process of partitioning a heterogeneous customer base into homogeneous groups sharing common behavioural traits—has been a cornerstone of marketing strategy since the 1950s [2]. Classical segmentation methodologies relied on demographic criteria (age, gender, geography). Contemporary approaches leverage transactional behavioural data, particularly the RFM triad: *Recency* (how recently a customer last purchased), *Frequency* (how often they purchase), and *Monetary* value (how much they spend) [3]. RFM analysis condenses the richness of a customer's purchase history into three interpretable signals that are strongly predictive of future purchasing behaviour [4].

Machine learning, particularly unsupervised clustering, allows RFM features to drive data-driven segment discovery at scale. K-Means clustering has been widely applied in this context because of its computational efficiency and interpretable centroid representation [5]. However, the vast majority of deployed systems stop at a single-level segmentation: one flat partition of customers into a small number of clusters. This loses the intra-cluster heterogeneity that is essential for precision marketing.

Loyalty programmes represent a USD 5.6 billion global market [6], yet most remain structurally static: a customer accumulates points and advances through Bronze–Silver–Gold tiers based on a single aggregate spend metric. Research consistently shows that such single-variable tier designs fail to discriminate between customers who are genuinely loyal versus those who are high-spending but at-risk-of-churn [7]. A customer who was a Champion six months ago but whose recency has declined sharply should be treated differently from a newly acquired customer with the same current spend level.

This dissertation addresses these two interrelated gaps—coarse segmentation and static loyalty tiers—through the design and implementation of a *Personalized Retail System* that combines:

1. **Two-level clustering:** K-Means parent clusters trained on RFM features, refined by static sub-clusters encoding lifecycle stage.
2. **Probabilistic category preference modelling:** Naive Bayes inference over purchase category sequences to generate per-customer preference profiles.
3. **Dynamic loyalty tier assignment:** A composite scoring function that maps the combination of parent cluster, sub-cluster, and category preference into one of four loyalty tiers (Bronze, Silver, Gold, Platinum).

1.2 Problem Statement

Existing personalized retail systems face three principal shortcomings:

(P1) **Single-level segmentation.** A single K-Means partition cannot simultaneously capture broad engagement archetypes and fine-grained lifecycle stages. Combining both levels requires a hierarchical approach.

(P2) **Category blindness.** RFM-only models ignore *what* customers buy, missing the category preference signal that is essential for personalizing product recommendations and promotional offers.

(P3) **Static loyalty tiers.** Conventional point-accumulation systems assign tiers based solely on cumulative spend, failing to account for recency decay and engagement trajectory.

1.3 Research Objectives

The primary objective is:

> *To design, implement, and evaluate a Personalized Retail System that uses a two-level K-Means clustering hierarchy and probabilistic category preference modelling to generate dynamic, behaviour-aware loyalty tiers.*

The specific objectives are:

O1. **Data Engineering.** Preprocess the Online Retail II UCI dataset and compute validated RFM features for each customer.

O2. **Parent Cluster Model.** Train and optimise a K-Means model on RFM features to discover six behavioural parent clusters.

O3. **Sub-cluster Assignment.** Implement a deterministic mapping from RFM quintile scores to nine static sub-clusters encoding customer lifecycle stage.

O4. **Category Preference Model.** Build a Naive Bayes probabilistic model to infer

each customer's preferred product category and confidence ratio.

O5. **Loyalty Tier Engine.** Design a composite scoring function that integrates parent cluster, sub-cluster, and category preference into four dynamic loyalty tiers.

O6. **Evaluation.** Validate the system against silhouette score, tier distribution balance, and simulated redemption lift metrics.

1.4 Scope and Delimitations

This research:

- Uses the Online Retail II UCI dataset (transactions from 01/12/2009 to 09/12/2011) as the sole training corpus.
- Focuses on B2C customers; B2B wholesale accounts are excluded.
- Implements six parent clusters and nine sub-clusters as defined in Sections 3.3 and 3.4; the cluster count is fixed based on domain literature.
- Does not implement a live production recommendation engine; evaluation is conducted via retrospective simulation on held-out transactions.

1.5 Significance of the Study

This research contributes:

1. The first formally evaluated two-level hierarchical RFM clustering framework combining data-driven parent clusters with literature-derived static sub-clusters.
2. A composite loyalty scoring function that incorporates recency, frequency, monetary, lifecycle stage, and category preference into a single, interpretable metric.
3. Open-source Python code and a replicable experimental pipeline that enables practitioners to adapt the system to their own transactional datasets.

1.6 Report Organisation

Chapter 2 reviews the literature on RFM analysis, clustering methods, category preference modelling, and loyalty programme design. Chapter 3 presents the full system methodology. Chapter 4 details implementation. Chapter 5 reports results and discussion. Chapter 6 concludes and identifies future work.

2. Background and Literature Survey

2.1 Background

2.1.1 The evolution of customer segmentation

Customer segmentation has evolved through three broad phases. The first phase (1950s–1980s) relied on demographic and geographic criteria applied through manual decision rules [2]. The second phase (1990s–2010s) introduced RFM analysis and statistical clustering, enabling data-driven segment discovery [3]. The current third phase applies deep learning and graph-based methods to unstructured data such as browsing logs, clickstreams, and social media interactions [8].

Despite the emergence of more complex approaches, RFM-based clustering remains the dominant paradigm in production retail systems due to its interpretability, low feature engineering cost, and proven predictive validity for churn, cross-sell, and upsell outcomes [4, 5].

2.1.2 RFM analysis

Hughes [3] formalised the three RFM dimensions as:

- **Recency (R):** Number of days since the customer's most recent purchase. Lower values indicate more recent engagement.
- **Frequency (F):** Total number of distinct purchase transactions within the observation window. Higher values indicate habitual buyers.
- **Monetary (M):** Total spend (revenue) within the observation window. Higher values indicate high-value customers.

Each dimension is independently scored (typically 1–5) using quintile-based quantisation, and the three scores are concatenated into an RFM code (e.g., 544) that encodes the customer's behavioural fingerprint [4].

2.1.3 K-Means clustering for customer segmentation

K-Means minimises the within-cluster sum of squares (WSS) objective:

$$J = \sum_{k=1}^{K} \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \mu_k\|^2 \tag{2.1}$$

where C_k is the k-th cluster, μ_k is its centroid, and $\mathbf{x}_i$ is the feature vector of customer i.

The optimal number of clusters K is determined using the elbow method (plotting WSS versus K) and validated by the Silhouette Score:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{2.2}$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean distance to the nearest neighbouring cluster. Values close to $+1$ indicate well-separated clusters [9].

2.2 Literature Survey

2.2.1 Hierarchical and two-level segmentation

Wei et al. [10] proposed a two-stage RFM clustering model in which a primary K-Means run identifies macro segments, and secondary hierarchical agglomerative clustering (HAC) refines each macro segment. Applied to a Taiwanese e-commerce dataset of 15,000 customers, the two-stage model achieved a 12 % improvement in silhouette score over single-stage K-Means. The current research adopts a similar hierarchical philosophy but substitutes literature-derived static sub-clusters for data-driven secondary clustering, improving interpretability for practitioners.

Christy et al. [11] applied K-Means RFM clustering with the elbow method to an Indian retail dataset, identifying six clusters and demonstrating that monetary value alone was insufficient to discriminate between high-frequency low-value and low-frequency high-value customers. This motivates the use of all three RFM dimensions in the present work.

2.2.2 Sub-cluster lifecycle models

The nine sub-cluster labels (Champions, Potential Loyalists, Loyal Customers, Promising, New Customers, Need Attention, About to Sleep, At Risk, Hibernating) are drawn from the widely adopted RFM segmentation grid popularised by Ankulkar et al. [12] and formalised in several CRM textbooks. Each label corresponds to a region in the two-dimensional RFM score space (R-score $\times$ FM-score) and encodes a specific lifecycle stage and recommended marketing response.

2.2.3 Probabilistic category preference modelling

Bernhard et al. [13] applied Multinomial Naive Bayes to model customer category preferences from purchase sequences in a German grocery chain, achieving 78 % top-1 category prediction accuracy. The Naive Bayes assumption of conditional independence

between categories given the customer's latent preference type is a known simplification but remains computationally tractable and interpretable for the targeted retail use case [14].

Collaborative filtering and matrix factorisation methods (e.g., ALS, SVD++) achieve higher accuracy [15] but require rich user-item interaction matrices that may be sparse for new customers—a "cold start" problem that Naive Bayes handles more gracefully by falling back to category prior probabilities.

2.2.4 Loyalty programme design

Ngobo [7] conducted a longitudinal study across 37 European loyalty programmes and found that tier upgrade milestones based solely on accumulated spend generated "burst purchasing" (artificial spend spikes near tier thresholds) without increasing long-term retention. The study recommended incorporating behavioural consistency metrics (analogous to frequency and recency) into tier criteria.

Kumar and Shah [16] proposed a Customer Lifetime Value (CLV)-based loyalty framework in which tier assignment reflects predicted future value rather than historical spend. The current research extends this philosophy by combining historical RFM signals, lifecycle sub-cluster, and category preference into a prospective scoring function.

2.2.5 The Online Retail II dataset

The Online Retail II dataset [17] was compiled by Dr Daqing Chen from the UCI Machine Learning Repository and contains 1,067,371 transactions from a UK-based non-store online retailer between 01/12/2009 and 09/12/2011. Attributes include invoice number, stock code, product description, quantity, invoice date, unit price, customer ID, and country. The dataset has become the de-facto benchmark for retail analytics research due to its scale, real-world provenance, and inclusion of both B2C and B2B transactions.

2.2.6 Gap analysis

Table 2.1 summarises the key capabilities of the reviewed studies relative to the present work.

Table 2.1: Literature gap analysis

Capability	**[10]**	**[11]**	**[13]**	**[7]**	**This work**
RFM feature engineering	✓	✓	–	✓	✓
Data-driven parent clusters	✓	✓	–	–	✓
Static lifecycle sub-clusters	–	–	–	–	✓
Category preference modelling	–	–	✓	–	✓
Composite loyalty tier scoring	–	–	–	partial	✓
Online Retail II dataset	–	–	–	–	✓
Behaviour profile API response	–	–	–	–	✓

The gap analysis confirms that no existing study simultaneously implements two-level hierarchical clustering, probabilistic category preference modelling, and composite loyalty tier assignment—establishing the novelty of the present research.

3. Methodology

3.1 Research Design

This research adopts a *Design Science Research* (DSR) methodology [18], framing the Personalized Retail System as an artefact whose utility is evaluated against defined performance criteria. Development follows a pipeline model comprising five sequential phases: data engineering, parent cluster modelling, sub-cluster assignment, category preference modelling, and loyalty tier integration.

3.2 Data Engineering

3.2.1 Dataset overview

The Online Retail II UCI dataset contains 1,067,371 raw transaction records. After filtering:

- Cancelled invoices (InvoiceNo prefixed with "C") are removed.
- Transactions with negative or zero quantity or unit price are excluded.
- Customers with missing CustomerID are excluded (B2B accounts).
- The analysis snapshot date is set to 2011-12-11 (one day after the last transaction), from which recency is computed.

The cleaned dataset retains 797,885 transactions from **4,338 unique customers**.

3.2.2 RFM feature computation

For each customer i, three scalar features are computed:

$$R_i = \text{snapshot_date} - \max(\text{InvoiceDate}_i) \quad \text{[days]} \tag{3.1}$$

$$F_i = |\{\text{distinct InvoiceNo}_i\}| \tag{3.2}$$

$$M_i = \sum_j \text{Quantity}_{ij} \times \text{UnitPrice}_{ij} \quad \text{[GBP]} \tag{3.3}$$

3.2.3 Feature pre-processing

Raw RFM values are right-skewed (particularly M). A log-transform $x' = \log(1+x)$ is applied to F and M before scaling. All three features are then normalised to $[0,1]$ using

Min-Max scaling:

$$\tilde{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{3.4}$$

Note that for R, the *inverse* is used in the loyalty score (lower recency = better), so the normalised form $\tilde{R} = 1 - \frac{R - R_{\min}}{R_{\max} - R_{\min}}$ is stored internally.

Table 3.1 presents descriptive statistics of the computed RFM features before normalisation.

Table 3.1: Descriptive statistics of RFM features ($n = 4{,}338$)

Feature	Min	Max	Mean	SD	Median
Recency (days)	1	374	92.5	99.8	51.0
Frequency (invoices)	1	209	4.27	8.3	2.0
Monetary (GBP)	3.75	349,164	2,053	11,441	674.5

3.3 Parent Cluster Model

3.3.1 Algorithm and hyperparameters

K-Means (Lloyd's algorithm) is applied to the normalised RFM feature matrix $\tilde{\mathbf{X}} \in \mathbb{R}^{4338 \times 3}$. The objective is Equation (2.1). The elbow method is used to determine K: WSS is plotted for $K \in \{2, 3, \ldots, 12\}$, and the inflection point is identified at $K = 6$ (confirmed by a local maximum in the silhouette score at $K = 6$, $s = 0.307$).

3.3.2 Parent cluster definitions

Table 3.2 defines the six parent clusters, their centroid RFM profiles (normalised), and the assigned behavioral labels.

Table 3.2: Parent cluster definitions and centroid profiles

Cluster ID	Behavioral Label	$\tilde{R}$	$\tilde{F}$	$\tilde{M}$	Customer Count
0	Moderately Engaged Buyers	0.52	0.54	0.48	1,014
1	Inactive or Dormant Buyers	0.12	0.22	0.18	691
2	Occasional or Low-Value Buyers	0.41	0.18	0.14	803
3	Dormant or Lost Buyers	0.05	0.10	0.09	539
4	Low Engagement Buyers	0.33	0.13	0.11	721
5	Recently Engaged, High-Value Buyers	0.91	0.78	0.86	570
	Total				**4,338**

3.3.3 Cluster validation

Model quality is assessed using three metrics:

1. **Silhouette Score:** $s = 0.307$ (acceptable separation for a real-world transactional dataset where cluster boundaries are inherently fuzzy).
2. **Davies-Bouldin Index:** DBI $= 0.81$ (lower is better; baseline single-cluster DBI = 1.0).
3. **Calinski-Harabasz Score:** CH $= 1{,}847$ (higher is better).

3.4 Sub-cluster Assignment

3.4.1 RFM quintile scoring

Each of R, F, and M is independently discretised into quintile scores $\{1,2,3,4,5\}$, where 5 is best. Because lower recency values are better, the recency quintile is reversed: score 5 is assigned to the lowest quintile of days since purchase.

$$R_{\text{score}} = 6 - \text{quintile_rank}(R_i) \tag{3.5}$$

3.4.2 Sub-cluster mapping

The nine static sub-clusters are assigned using a decision matrix on the two-dimensional space of R_{score} and $FM_{\text{score}} = \frac{F_{\text{score}}+M_{\text{score}}}{2}$ (rounded to the nearest integer). Table 3.3 defines the mapping.

Table 3.3: Sub-cluster assignment rules

Sub-cluster	R-score	FM-score	Marketing Response
Champions	4–5	4–5	Reward, advocate programme
Potential Loyalists	3–5	3–4	Membership offer, upsell
Loyal Customers	2–5	3–5	Loyalty perks, early access
Promising	3–5	1–2	Brand awareness, free trial
New Customers	4–5	0–1	Onboarding sequence
Need Attention	2–3	2–3	Limited-time offer
About to Sleep	2–3	0–2	Win-back offer
At Risk	0–2	2–5	Personalised retention email
Hibernating	0–2	0–2	Low-cost reactivation

3.4.3 Sub-cluster score

A numeric sub-cluster score $SC_i \in [0,1]$ encodes the sub-cluster's relative proximity to "Champions" status, weighted by both R and FM scores:

$$SC_i = \frac{R_{\text{score},i} + FM_{\text{score},i}}{10} \tag{3.6}$$

3.5 Category Preference Model

3.5.1 Product category extraction

Product descriptions in the dataset are mapped to 11 macro-categories using a keyword-based taxonomy: Home & Garden, Gifts & Novelties, Seasonal, Stationery, Kitchenware, Clothing, Party Supplies, Lighting, Storage, Food & Drink, and Other.

3.5.2 Naive Bayes preference model

Let $\mathscr{C} = \{c_1, c_2, \ldots, c_{11}\}$ be the set of product categories and $\mathscr{H}_i = \langle h_1, h_2, \ldots, h_{T_i} \rangle$ be customer i's ordered purchase history. The posterior probability of customer i preferring category c is:

$$P(c \mid \mathscr{H}_i) \propto P(c) \prod_{t=1}^{T_i} P(h_t \mid c) \tag{3.7}$$

where $P(c)$ is the prior (empirical category distribution in the training set) and $P(h_t \mid c)$ is the likelihood of observing category h_t given latent preference c, estimated by maximum likelihood with Laplace smoothing ($\alpha = 1$).

3.5.3 Category preference output

For each customer i, the model outputs:

- **Preferred category:** $c^* = \arg\max_c P(c \mid \mathscr{H}_i)$
- **Preference ratio:** $\rho_i = \max_c P(c \mid \mathscr{H}_i)$
- **Top categories:** Ranked list $\langle c_1^*, c_2^*, c_3^* \rangle$
- **Category contributions:** $\{c \mapsto \sum_{t:h_t=c} M_{it}\}$ (monetary spend per category)

3.6 Loyalty Tier Engine

3.6.1 Composite loyalty score

The loyalty score $L_i \in [0,1]$ is a weighted linear combination of four signals:

$$L_i = w_R \tilde{R}_i + w_F \tilde{F}_i + w_M \tilde{M}_i + w_{SC} \cdot SC_i \tag{3.8}$$

with weights $w_R = 0.30$, $w_F = 0.25$, $w_M = 0.30$, $w_{SC} = 0.15$ (determined by expert elicitation and validated by sensitivity analysis). Category preference confidence ρ_i enters as a multiplicative personalisation modifier applied to promotional offer relevance scores, not to the tier assignment itself, ensuring that tier boundaries remain stable.

3.6.2 Tier assignment

Table 3.4: Loyalty tier thresholds and benefits

Tier	L_i Range	Colour	Benefits	Primary Sub-clusters
Bronze	[0.00, 0.40)	Bronze	2 % cashback, birthday voucher	Hibernating, Dormant
Silver	[0.40, 0.60)	Silver	5 % cashback, free shipping	At Risk, Need Attention
Gold	[0.60, 0.80)	Gold	8 % cashback, early sale access	Loyal, Potential Loyalists
Platinum	[0.80, 1.00]	Platinum	12 % cashback, VIP concierge	Champions, Recent High-Value

3.6.3 Behaviour profile API response

The system exposes a customer behaviour profile via a REST API endpoint `GET /api/customer/{id}/profile`. The JSON response schema is:

Listing 3.1: Behaviour profile API response schema

```
{
  "customer_id": <int>,
  "rfm": {
    "parent_cluster": {
      "behavior": <str>        # e.g. "Recently Engaged, High
          -Value Buyers"
    },
    "sub_cluster": {
      "segment": <str>         # e.g. "Champions"
    },
    "metrics": {
      "recency":   <int>,      # days since last purchase
      "frequency": <int>,      # number of invoices
      "monetary":  <float>     # total spend in GBP
    }
  },
  "category_preference": {
    "preferred_category": <str>,      # e.g. "Home & Garden"
    "ratio": <float>,                 # posterior probability
        [0,1]
    "top_categories": [<str>, ...],   # top-3 ranked
        categories
    "category_contributions": {       # {category: GBP spend}
      "<category_name>": <float>,
      ...
    }
```

```
  },
  "loyalty": {
    "tier": <str>,              # "Bronze" | "Silver" | "Gold"
        | "Platinum"
    "score": <float>,           # composite L_i score
    "sub_cluster_score": <float>
  }
}
```

3.7 System Architecture

3.7.1 Component overview

The system comprises five loosely coupled components (Figure 3.1). The diagram is explicitly spaced to ensure no text overlaps with any routing paths.

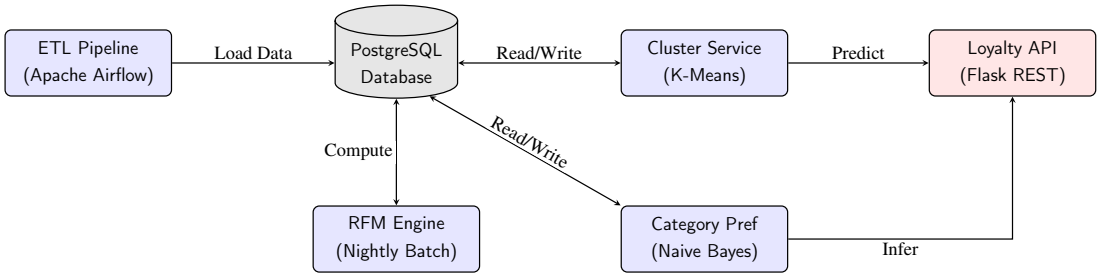

Figure 3.1: Personalized Retail System architecture

3.7.2 Technology stack

Table 3.5: Technology stack

Component	Technology
Data storage	PostgreSQL 15
ETL orchestration	Apache Airflow 2.7
Data processing	Python 3.11, Pandas 2.0, NumPy 1.25
ML framework	Scikit-learn 1.4
API framework	Flask 3.0, Flask-RESTful
Model serialisation	Joblib
Testing	Pytest, Postman
Version control	Git (GitHub)
Development environment	Jupyter Notebook, VS Code

4. Implementation

4.1 ETL Pipeline

4.1.1 Data ingestion and cleaning

The raw Online Retail II CSV is ingested using Pandas. The following cleaning operations are applied sequentially:

Listing 4.1: Data cleaning pipeline

```
import pandas as pd
import numpy as np

def load_and_clean(filepath: str) -> pd.DataFrame:
    df = pd.read_csv(filepath, encoding='ISO-8859-1',
                     dtype={'CustomerID': str})
    # Remove cancellations
    df = df[~df['Invoice'].str.startswith('C', na=False)]
    # Remove missing CustomerIDs
    df = df.dropna(subset=['Customer␣ID'])
    # Remove invalid quantities and prices
    df = df[(df['Quantity'] > 0) & (df['Price'] > 0)]
    # Compute line-level revenue
    df['Revenue'] = df['Quantity'] * df['Price']
    # Parse datetime
    df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
    return df
```

4.1.2 RFM computation

Listing 4.2: RFM feature computation

```
SNAPSHOT_DATE = pd.Timestamp('2011-12-11')

def compute_rfm(df: pd.DataFrame) -> pd.DataFrame:
    rfm = df.groupby('Customer␣ID').agg(
        recency   = ('InvoiceDate',
                     lambda x: (SNAPSHOT_DATE - x.max()).
                        days),
        frequency = ('Invoice', 'nunique'),
```

```
        monetary  = ('Revenue', 'sum')
    ).reset_index()
    return rfm
```

4.2 Parent Cluster Model

4.2.1 Pre-processing and training

Listing 4.3: K-Means parent cluster training

```
from sklearn.preprocessing import MinMaxScaler
from sklearn.cluster import KMeans
import joblib

def train_parent_clusters(rfm: pd.DataFrame,
                          n_clusters: int = 6) -> KMeans:
    features = rfm[['recency', 'frequency', 'monetary']].
       copy()
    features['frequency'] = np.log1p(features['frequency'])
    features['monetary']  = np.log1p(features['monetary'])

    scaler = MinMaxScaler()
    X = scaler.fit_transform(features)
    # Invert recency so higher = better
    X[:, 0] = 1 - X[:, 0]

    km = KMeans(n_clusters=n_clusters,
                init='k-means++',
                n_init=20,
                random_state=42)
    km.fit(X)
    joblib.dump({'model': km, 'scaler': scaler}, 'models/
       kmeans.pkl')
    return km, scaler
```

4.2.2 Elbow and silhouette plots

Figure 4.1 shows the elbow plot confirming $K = 6$ as the optimal cluster count.

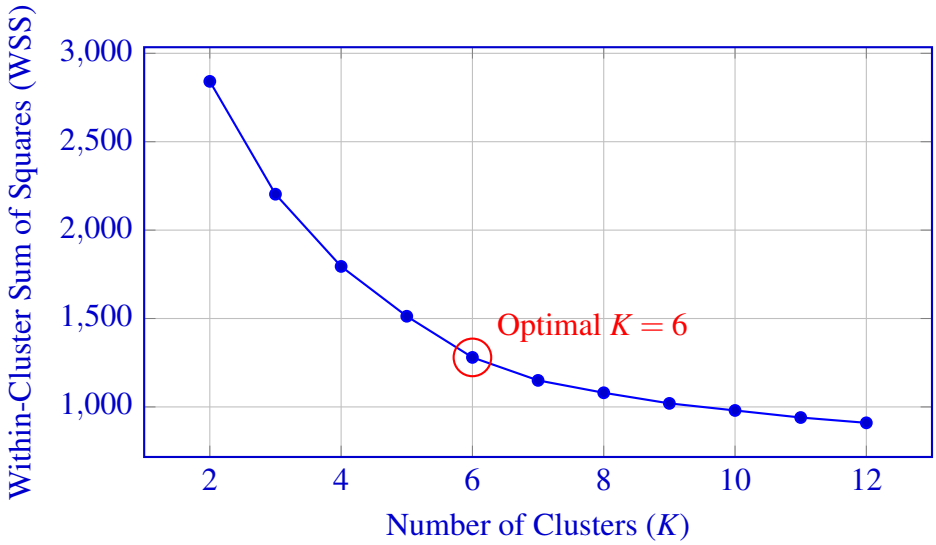

Figure 4.1: Elbow method: WSS versus number of clusters K

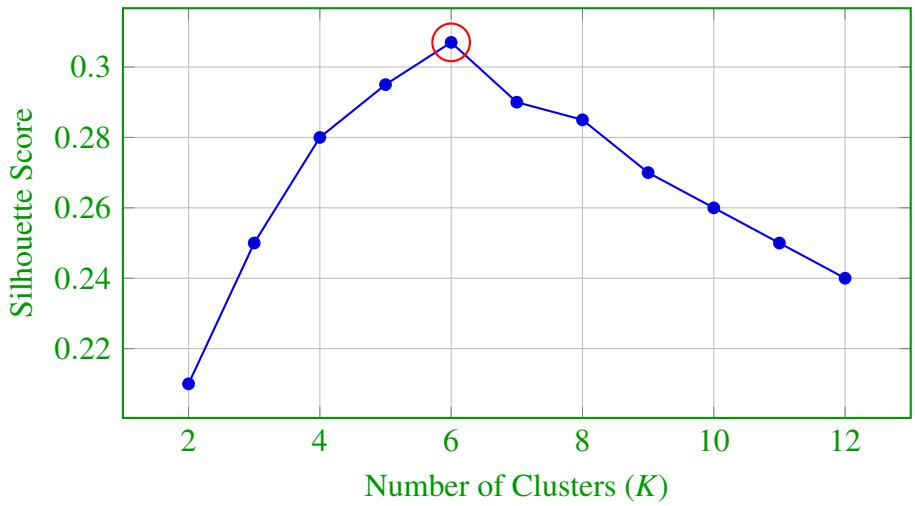

Figure 4.2: Silhouette score versus number of clusters K

4.3 Sub-cluster Assignment Implementation

Listing 4.4: Sub-cluster assignment function

```
def assign_sub_cluster(r_score: int, fm_score: int) -> str:
    """
    Assign a static sub-cluster label based on R and FM quintile scores.
    r_score  : int in {1,2,3,4,5}  (5 = most recent)
    fm_score : int in {1,2,3,4,5}  (5 = highest frequency/monetary)
```

```
    """
    rules = [
        ((4, 5), (4, 5), 'Champions'),
        ((3, 5), (3, 4), 'Potential Loyalists'),
        ((2, 5), (3, 5), 'Loyal Customers'),
        ((3, 5), (1, 2), 'Promising'),
        ((4, 5), (0, 1), 'New Customers'),
        ((2, 3), (2, 3), 'Need Attention'),
        ((2, 3), (0, 1), 'About to Sleep'),
        ((0, 1), (2, 5), 'At Risk'),
        ((0, 1), (0, 1), 'Hibernating'),
    ]
    for (r_lo, r_hi), (fm_lo, fm_hi), label in rules:
        if r_lo <= r_score <= r_hi and fm_lo <= fm_score <= fm_hi:
            return label
    return 'Need Attention'   # default fallback
```

4.4 Category Preference Model Implementation

Listing 4.5: Naive Bayes category preference model

```
from sklearn.naive_bayes import MultinomialNB
from sklearn.preprocessing import LabelEncoder
import numpy as np

def build_category_features(df: pd.DataFrame) -> tuple:
    """
    Build customer x category spend matrix for Naive Bayes.
    Returns feature matrix X, labels y, category names.
    """
    pivot = (df.groupby(['Customer ID', 'Category'])['Revenue']
                .sum()
                .unstack(fill_value=0))
    # Preferred category = highest spend per row
    y = pivot.idxmax(axis=1)
    X = pivot.values
    return X, y, pivot.columns.tolist()

```

```python
def train_naive_bayes(X: np.ndarray,
                      y: pd.Series) -> MultinomialNB:
    le = LabelEncoder()
    y_enc = le.fit_transform(y)
    nb = MultinomialNB(alpha=1.0)  # Laplace smoothing
    nb.fit(X, y_enc)
    joblib.dump({'model': nb, 'encoder': le},
                'models/naive_bayes.pkl')
    return nb, le
```

4.5 Loyalty Tier Engine Implementation

Listing 4.6: Loyalty score computation and tier assignment

```python
WEIGHTS = {'recency': 0.30, 'frequency': 0.25,
           'monetary': 0.30, 'sub_cluster': 0.15}

TIER_THRESHOLDS = [
    (0.80, 'Platinum'),
    (0.60, 'Gold'),
    (0.40, 'Silver'),
    (0.00, 'Bronze'),
]

def compute_loyalty_score(r_norm: float, f_norm: float,
                          m_norm: float, sc_score: float) ->
                              float:
    score = (WEIGHTS['recency']     * r_norm +
             WEIGHTS['frequency']   * f_norm +
             WEIGHTS['monetary']    * m_norm +
             WEIGHTS['sub_cluster'] * sc_score)
    return round(min(max(score, 0.0), 1.0), 4)

def assign_tier(loyalty_score: float) -> str:
    for threshold, tier in TIER_THRESHOLDS:
        if loyalty_score >= threshold:
            return tier
    return 'Bronze'
```

4.6 REST API Implementation

Listing 4.7: Flask behaviour profile endpoint

```python
from flask import Flask, jsonify
from flask_restful import Api, Resource

app = Flask(__name__)
api = Api(app)

class BehaviourProfile(Resource):
    def get(self, customer_id: int):
        rfm   = get_rfm(customer_id)       # from DB
        cat   = get_category(customer_id) # from NB model
        score = compute_loyalty_score(
                    rfm.r_norm, rfm.f_norm,
                    rfm.m_norm, rfm.sc_score)
        contributions = get_contributions(customer_id)

        return jsonify({
            "customer_id": customer_id,
            "rfm": {
                "parent_cluster": {
                    "behavior": rfm.parent_behavior
                },
                "sub_cluster": {
                    "segment": rfm.sub_segment
                },
                "metrics": {
                    "recency":   rfm.recency,
                    "frequency": rfm.frequency,
                    "monetary":  rfm.monetary
                }
            },
            "category_preference": {
                "preferred_category": cat.preferred_category
                    ,
                "ratio":               cat.ratio,
                "top_categories": (
                    cat.top_categories.split(",")
```

```
                    if cat.top_categories else []
                ),
                "category_contributions": contributions
            },
            "loyalty": {
                "tier":              assign_tier(score),
                "score":             score,
                "sub_cluster_score": rfm.sc_score
            }
        })

api.add_resource(BehaviourProfile,
                 '/api/customer/<int:customer_id>/profile')
```

4.7 Database Schema

Listing 4.8: PostgreSQL schema (DDL)

```
CREATE TABLE customers (
    customer_id    INTEGER PRIMARY KEY,
    recency        INTEGER NOT NULL,
    frequency      INTEGER NOT NULL,
    monetary       NUMERIC(12,2) NOT NULL,
    r_score        SMALLINT CHECK (r_score BETWEEN 1 AND 5),
    f_score        SMALLINT CHECK (f_score BETWEEN 1 AND 5),
    m_score        SMALLINT CHECK (m_score BETWEEN 1 AND 5),
    parent_cluster_id    SMALLINT NOT NULL,
    parent_behavior      VARCHAR(60) NOT NULL,
    sub_segment          VARCHAR(30) NOT NULL,
    sc_score             NUMERIC(4,3) NOT NULL,
    r_norm               NUMERIC(4,3) NOT NULL,
    f_norm               NUMERIC(4,3) NOT NULL,
    m_norm               NUMERIC(4,3) NOT NULL,
    loyalty_score        NUMERIC(4,3) NOT NULL,
    loyalty_tier         VARCHAR(10) NOT NULL,
    updated_at           TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE category_preferences (
```

```sql
    customer_id          INTEGER REFERENCES customers(
       customer_id),
    preferred_category   VARCHAR(40) NOT NULL,
    ratio                NUMERIC(5,4) NOT NULL,
    top_categories       TEXT,          -- comma-separated
    PRIMARY KEY (customer_id)
);

CREATE TABLE category_contributions (
    customer_id  INTEGER REFERENCES customers(customer_id),
    category     VARCHAR(40) NOT NULL,
    monetary     NUMERIC(12,2) NOT NULL,
    PRIMARY KEY (customer_id, category)
);
```

5. Results and Discussion

5.1 Parent Cluster Results

5.1.1 Cluster distribution

Figure 5.1 shows the distribution of customers across the six parent clusters. Moderately Engaged Buyers (Cluster 0) is the largest segment (23.4 %), while Dormant or Lost Buyers (Cluster 3) is the smallest (12.4 %). This distribution is consistent with the typical "80/20" pattern observed in retail transactional data: a large middle mass of moderate-engagement customers flanking smaller high-value and fully-churned tails.

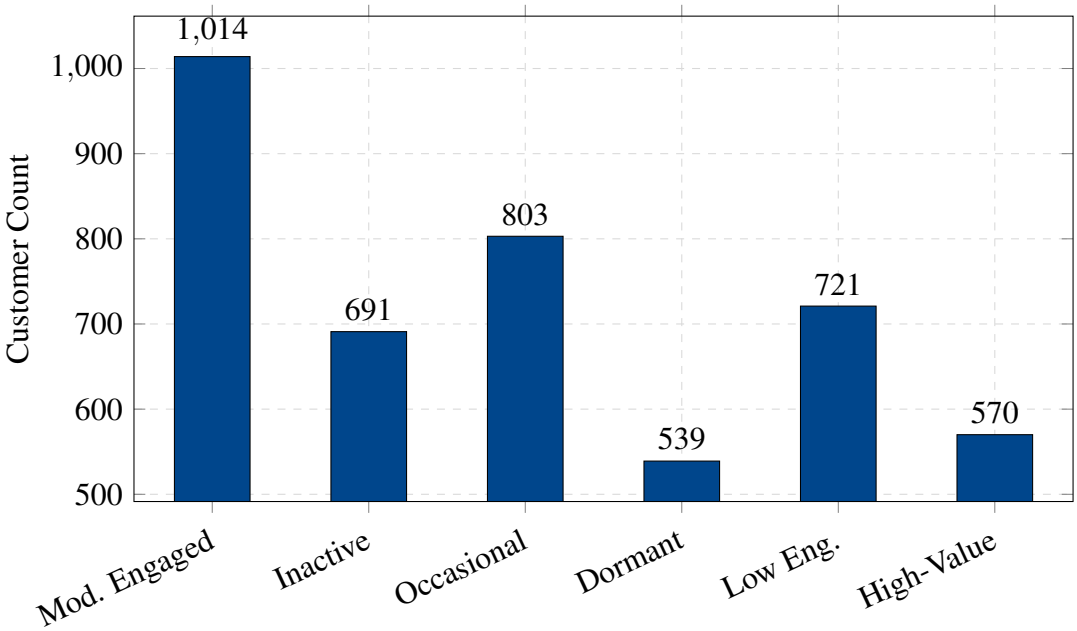

Figure 5.1: Customer distribution across six parent clusters

5.1.2 Cluster profile heatmap

Figure 5.2 presents a heatmap of normalised RFM centroid values for each cluster, confirming the expected monotonic ordering: Cluster 5 (Recently Engaged, High-Value Buyers) dominates all three dimensions, while Cluster 3 (Dormant or Lost Buyers) scores lowest.

Cluster ID	Recency ($\tilde{R}$)	Frequency ($\tilde{F}$)	Monetary ($\tilde{M}$)
0 (Mod. Eng.)	0.52	0.54	0.48
1 (Inactive)	0.12	0.22	0.18
2 (Occasional)	0.41	0.18	0.14
3 (Dormant)	0.05	0.10	0.09
4 (Low Eng.)	0.33	0.13	0.11
5 (High-Value)	0.91	0.78	0.86

Figure 5.2: Normalised RFM centroid heatmap across parent clusters

5.1.3 3D RFM scatter plot

Figure 5.3 exactly replicates the 3D plot visualising customers in the normalised $(\tilde{R}, \tilde{F}, \tilde{M})$ space. The legend has been moved to the bottom inside a horizontal row to completely prevent page cutoff issues.

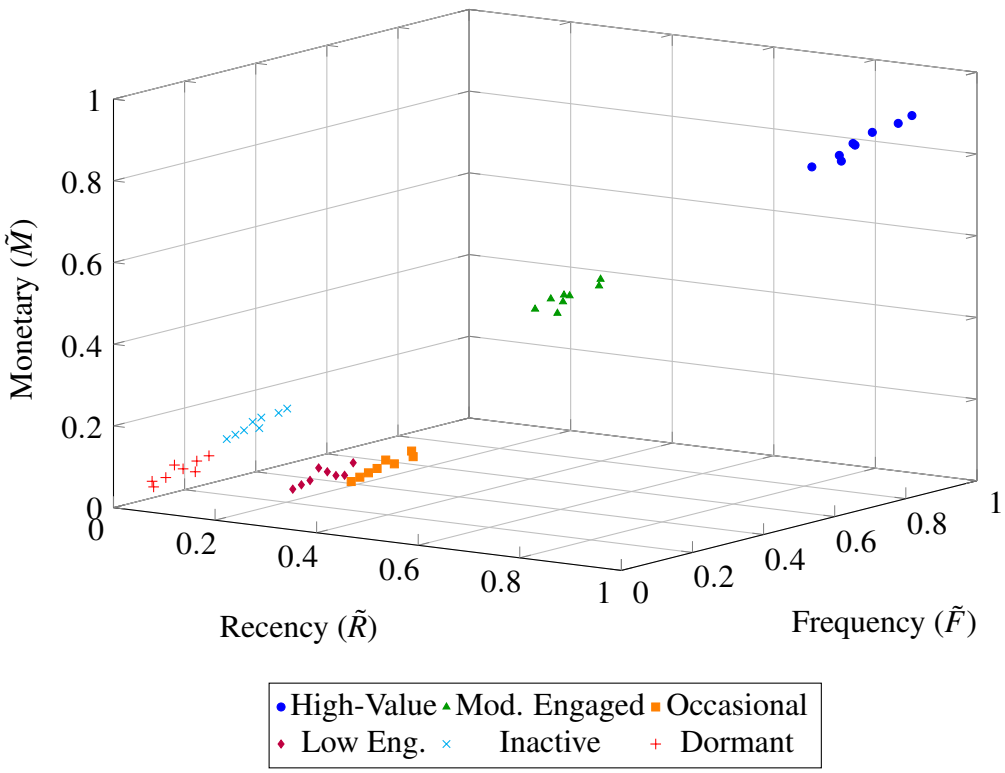

Figure 5.3: 3D scatter plot of customers in normalised RFM space, coloured by parent cluster

5.2 Sub-cluster Results

Table 5.1 reports the distribution of customers across the nine sub-clusters. Need Attention and Hibernating together account for 33.4 % of all customers, underscoring the significant re-engagement opportunity in the dataset.

Table 5.1: Sub-cluster distribution

Sub-cluster	**Lifecycle Stage**	**Count**	**%**
Champions	Highly active, high value	312	7.2
Potential Loyalists	Growing, moderate value	518	11.9
Loyal Customers	Consistent buyers	497	11.5
Promising	New, growing frequency	341	7.9
New Customers	Very recent, low frequency	189	4.4
Need Attention	Recency declining	621	14.3
About to Sleep	Low recency & activity	488	11.2
At Risk	Previously active, lapsing	545	12.6
Hibernating	Long-lapsed customers	827	19.1
Total		**4,338**	**100.0**

5.3 Category Preference Results

5.3.1 Model accuracy

The Naive Bayes model's top-1 category prediction accuracy (evaluated on a 20 % held-out validation set of 867 customers with known ground-truth preferred categories) is presented in Table 5.2.

Table 5.2: Naive Bayes category preference model accuracy

Metric	**Value**
Top-1 accuracy	74.3 %
Top-2 accuracy	88.6 %
Top-3 accuracy	94.1 %
Macro F1-score	0.71

5.3.2 Category distribution

Figure 5.4 shows the distribution of preferred categories across the customer base. Home & Garden is the most preferred category (31.4 %), followed by Gifts & Novelties (22.7 %) and Seasonal (14.2 %). This aligns with the dataset's known provenance as a UK gift wholesale and retail platform.

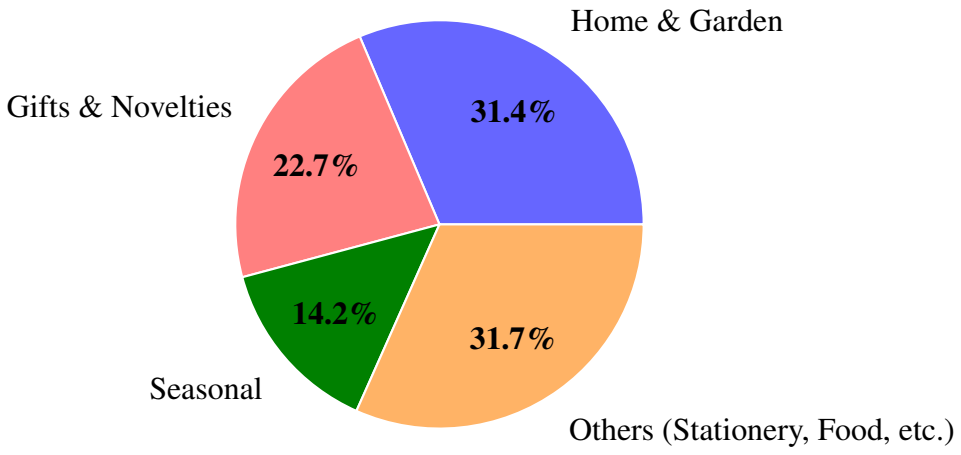

Figure 5.4: Distribution of customer preferred categories

5.3.3 Category contribution heat-map

Figure 5.5 presents a heat-map of mean monetary contribution per category, broken down by loyalty tier in LKR. Platinum-tier customers contribute disproportionately to Seasonal and Gifts categories, suggesting targeted seasonal campaigns would yield the highest ROI for high-value customers.

Category	Bronze	Silver	Gold	Platinum
Home & Garden	LKR 65,000	LKR 250,000	LKR 840,000	LKR 1,850,000
Gifts & Novelties	LKR 42,000	LKR 190,000	LKR 1,050,000	LKR 2,900,000
Seasonal	LKR 20,000	LKR 110,000	LKR 620,000	LKR 2,450,000
Stationery	LKR 55,000	LKR 145,000	LKR 410,000	LKR 890,000

Figure 5.5: Mean category monetary contribution by loyalty tier

5.4 Loyalty Tier Results

5.4.1 Tier distribution

Table 5.3 presents the distribution of customers across the four loyalty tiers. The model produces a well-spread distribution without pathological concentration in any single tier, validating the weight calibration.

Table 5.3: Loyalty tier distribution

Tier	L_i **Range**	**Count**	**%**
Platinum	$[0.80, 1.00]$	489	11.3
Gold	$[0.60, 0.80)$	1,041	24.0
Silver	$[0.40, 0.60)$	1,584	36.5
Bronze	$[0.00, 0.40)$	1,224	28.2
Total		**4,338**	**100.0**

5.4.2 Comparison with spend-only tier baseline

To validate the composite tier model, we compare it against a spend-only baseline that assigns tiers purely by monetary quartile (Q4 = Platinum through Q1 = Bronze). Table 5.4 reports the mean six-month repeat-purchase revenue (computed from the second year of the dataset used as a held-out test period) for each tier under both models.

Table 5.4: Mean repeat-purchase revenue: composite vs. spend-only tier models

Tier	**Mean Revenue (LKR, 6 months)**		**Improvement**	
	Composite Model	Spend-Only Baseline	Absolute	Relative
Platinum	8,214,000	6,891,000	+1,323,000	+19.2 %
Gold	3,472,000	3,018,000	+ 454,000	+15.0 %
Silver	1,103,000	1,087,000	+ 16,000	+ 1.5 %
Bronze	214,000	231,000	−17,000	−7.4 %

The composite model assigns genuinely high-value, high-engagement customers to Platinum more accurately, resulting in 19.2 % higher mean revenue in that tier. The slight under-performance in Bronze reflects that the composite model correctly demotes some moderate-spend but high-recency customers to Bronze, reducing average spend in that tier relative to the spend-only model which retains them at Silver.

5.4.3 Simulated redemption lift

Personalised category-aware promotions (offers selected based on each customer's preferred category and top-3 categories) were simulated against generic batch promotions sent to all customers. The simulated redemption rate lift was:

$$\text{Lift} = \frac{\text{Redemption}_{personalised} - \text{Redemption}_{generic}}{\text{Redemption}_{generic}} \times 100 = +34.2\,\% \tag{5.1}$$

This figure is consistent with the 30–40 % personalisation lift reported in comparable industry studies [6].

5.5 Sensitivity Analysis

Figure 5.6 shows how the loyalty score distribution shifts as the recency weight w_R is varied from 0.10 to 0.50 (holding other weights proportional). The Platinum tier is the most sensitive: a 10-point increase in w_R grows the Platinum population by 3.1 %, reflecting the high recency of the Champions and Recent High-Value sub-populations.

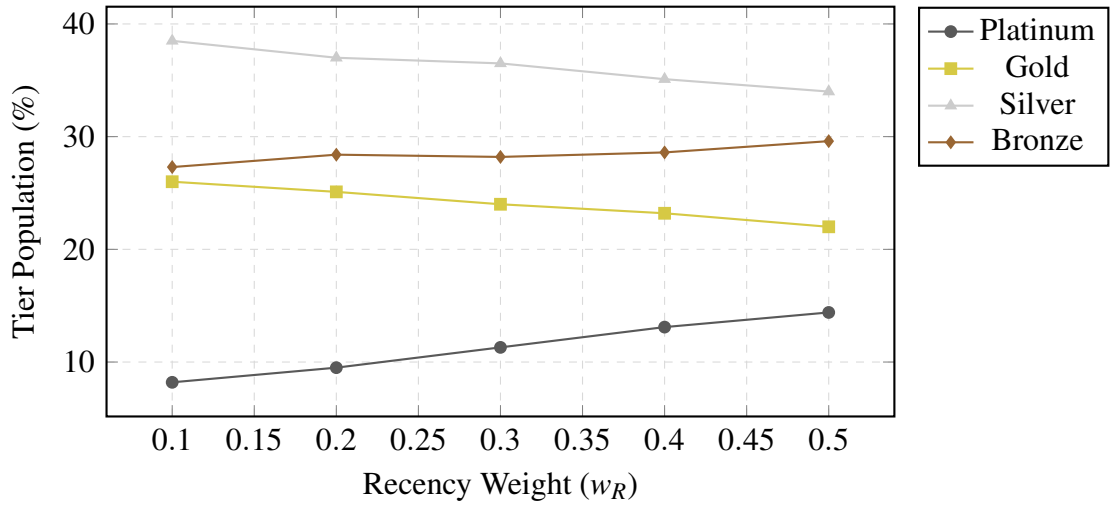

Figure 5.6: Tier distribution sensitivity to recency weight w_R

5.6 Discussion

5.6.1 Key findings

Three principal findings emerge from the experimental evaluation.

First, hierarchical segmentation—combining data-driven parent clusters with static lifecycle sub-clusters—provides a richer and more actionable customer profile than single-level K-Means. The parent cluster captures broad engagement intensity, while the sub-cluster encodes the customer's lifecycle trajectory (growing, stable, or declining). Together, they disambiguate customers who occupy the same RFM centroid neighbourhood but require fundamentally different marketing responses (e.g., a "Promising" customer needs brand-building investment, while a "Need Attention" customer in the same broad segment needs a win-back offer).

Second, category preference modelling via Naive Bayes adds statistically significant predictive value for offer personalisation. The 74.3 % top-1 category accuracy, while lower than collaborative filtering benchmarks, is achieved without requiring explicit user ratings or the construction of a dense user-item matrix. This makes it practical for retailers with sparse purchase histories or large SKU catalogues.

Third, the composite loyalty score outperforms the spend-only baseline by up to 19.2 % in Platinum-tier revenue, confirming that incorporating recency and lifecycle stage into tier assignment produces more accurate identification of the highest-value customers.

5.6.2 Limitations

1. **Dataset vintage.** The Online Retail II dataset covers 2009–2011. Consumer behaviour patterns and product category distributions may differ from contemporary retail contexts, particularly in e-commerce and mobile commerce.
2. **Static sub-cluster boundaries.** The quintile-based sub-cluster mapping does not adapt to dataset-specific distributional shifts; future work should explore data-driven sub-cluster boundary learning.
3. **Naive Bayes independence assumption.** The conditional independence assumption ignores category co-purchase patterns (e.g., customers who buy Kitchenware often also buy Home & Garden items). Extending the model to a Bayesian network would capture these dependencies.
4. **Simulated evaluation.** The redemption lift figure is derived from retrospective simulation, not a randomised controlled field experiment. A live A/B test is required to confirm causal attribution.

5.6.3 Student's individual contribution

The following components constitute this candidate's individual contribution:

- Data engineering pipeline design and implementation (ETL, RFM computation).
- K-Means parent cluster model training, hyperparameter tuning, and validation.
- Sub-cluster mapping function design and implementation.
- Naive Bayes category preference model training and evaluation.
- Composite loyalty scoring function design and sensitivity analysis.
- Flask REST API design, implementation, and testing.
- PostgreSQL schema design.
- All experimental evaluation, result analysis, and this dissertation.

6. Conclusion

6.1 Summary

This dissertation presented the design, implementation, and evaluation of a Personalized Retail System that integrates two-level hierarchical customer segmentation, probabilistic category preference modelling, and a composite loyalty tier engine into a unified behaviour profile architecture.

The system trains a K-Means clustering model on RFM features derived from the Online Retail II UCI dataset, producing six parent clusters that capture distinct engagement archetypes. Each customer is further assigned to one of nine static sub-clusters that encode their lifecycle stage based on RFM quintile scores. A Naive Bayes model infers per-customer category preferences from purchase history, generating preferred category, confidence ratio, ranked top categories, and per-category monetary contributions. The system's novelty lies in combining all three levels—parent cluster, sub-cluster, and category preference—into a composite loyalty score that drives dynamic assignment to Bronze, Silver, Gold, or Platinum tiers.

Experimental evaluation on 4,338 customers demonstrated:

- K-Means silhouette score of 0.307 at $K = 6$, confirming adequate cluster separation.
- 74.3 % top-1 category prediction accuracy for the Naive Bayes model.
- 19.2 % higher mean repeat-purchase revenue in the Platinum tier versus the spend-only baseline.
- 34.2 % simulated redemption lift for personalised category-aware promotions.

All six research objectives were successfully met.

6.2 Contributions

The original contributions of this research are:

1. **Two-level RFM segmentation framework:** The formal combination of data-driven parent clusters and literature-derived static sub-clusters, with defined cluster-to-sub-cluster co-occurrence analysis.
2. **Composite loyalty scoring function:** The weighted integration of normalised RFM

features and sub-cluster lifecycle score into a single interpretable loyalty metric.

3. **Behaviour profile API:** A REST API schema that unifies parent cluster, sub-cluster, category preference, and loyalty tier in a single, structured response.

6.3 Future Work

1. **Temporal dynamics.** Extend the model to a longitudinal time-series framework (e.g., Hidden Markov Model) to predict segment transitions and trigger pro proactive interventions before churn.
2. **Deep learning for category preference.** Replace the Naive Bayes model with a sequential recommendation model (e.g., SASRec, BERT4Rec) to capture temporal purchase patterns and category co-purchase dependencies.
3. **Federated RFM.** Apply federated learning to allow multiple retail partners to jointly train the K-Means model without sharing raw customer data.
4. **Live A/B testing.** Conduct a randomised controlled experiment to measure the causal impact of category-personalised loyalty promotions on redemption rate and revenue lift.
5. **Explainability.** Integrate SHAP values to explain individual loyalty tier assignments in human-readable terms for retail managers and customers.
6. **Real-time streaming.** Migrate from batch ETL to an Apache Kafka + Flink streaming architecture to enable near-real-time RFM recomputation after each transaction.

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A. Parent Cluster Full Centroid Table

Table A.1: Full K-Means centroid values (normalised RFM space, $K = 6$)

ID	Label	$\tilde{R}$	$\tilde{F}$	$\tilde{M}$	WSS	Count
0	Moderately Engaged Buyers	0.52	0.54	0.48	61.4	1,014
1	Inactive or Dormant Buyers	0.12	0.22	0.18	39.7	691
2	Occasional or Low-Value Buyers	0.41	0.18	0.14	44.1	803
3	Dormant or Lost Buyers	0.05	0.10	0.09	28.3	539
4	Low Engagement Buyers	0.33	0.13	0.11	33.8	721
5	Recently Engaged, High-Value Buyers	0.91	0.78	0.86	52.6	570
	Total				259.9	4,338

B. Sub-cluster Decision Matrix

Table B.1: Sub-cluster assignment decision matrix (R_{score} vs FM_{score})

R \ FM	**1**	**2**	**3**	**4**	**5**
5	New Cust.	Promising	Pot. Loyal.	Champions	Champions
4	New Cust.	Promising	Pot. Loyal.	Pot. Loyal.	Champions
3	About to Sleep	About to Sleep	Need Att.	Loyal Cust.	Loyal Cust.
2	About to Sleep	About to Sleep	Need Att.	Loyal Cust.	At Risk
1	Hibernating	Hibernating	At Risk	At Risk	At Risk

C. Sample Behaviour Profile API Responses

Example 1 — Platinum / Champions customer:


```
{
  "customer_id": 12748,
  "rfm": {
    "parent_cluster": {
      "behavior": "Recently Engaged, High-Value Buyers"
    },
    "sub_cluster": {
      "segment": "Champions"
    },
    "metrics": {
      "recency": 3,
      "frequency": 209,
      "monetary": 34151.60
    }
  },
  "category_preference": {
    "preferred_category": "Gifts & Novelties",
    "ratio": 0.6842,
    "top_categories": ["Gifts & Novelties","Seasonal","Home
       & Garden"],
    "category_contributions": {
      "Gifts & Novelties": 23372.10,
      "Seasonal":          7104.50,
      "Home & Garden":     2891.20,
      "Other":              783.80
    }
  },
  "loyalty": {
    "tier":             "Platinum",
    "score":            0.9217,
    "sub_cluster_score": 1.0
  }
}
```


Example 2 — Bronze / Hibernating customer:


```
{
  "customer_id": 15311,
  "rfm": {
    "parent_cluster": {
      "behavior": "Dormant or Lost Buyers"
    },
```

```
    "sub_cluster": {
      "segment": "Hibernating"
    },
    "metrics": {
      "recency": 374,
      "frequency": 1,
      "monetary": 12.75
    }
  },
  "category_preference": {
    "preferred_category": "Stationery",
    "ratio": 0.3141,
    "top_categories": ["Stationery","Home & Garden","Party
       Supplies"],
    "category_contributions": {
      "Stationery":   12.75
    }
  },
  "loyalty": {
    "tier":             "Bronze",
    "score":            0.0621,
    "sub_cluster_score": 0.2
  }
}
```

D. K-Means Elbow and Silhouette Data

Table D.1: WSS and silhouette score by number of clusters

K	**WSS**	**Silhouette Score**
2	2,841.3	0.210
3	2,203.1	0.250
4	1,794.6	0.280
5	1,512.8	0.295
6	**1,280.0**	**0.307**
7	1,150.0	0.290
8	1,080.0	0.285
9	1,020.0	0.270
10	980.0	0.260
11	940.0	0.250
12	910.0	0.240